

Supporting Non-Musicians' Creative Engagement with “EasyPlay!”, a music production software

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Abstract— This project develops a music production software with the goal of guiding and supporting non-musicians in their creative engagement. The software design incorporates generators, a graphical score-based sequencer, and a user-friendly interface following a flat design approach. A survey and evaluations were conducted to gather user preferences and assess the software's impact. The results show that the software facilitates creative expression, offering guidance and allowing users to experiment with music composition. Further research is needed to refine the design and validate its effectiveness across a wider user base.

Author Keywords—Digital Audio Workstation, Flat Design, Creative Engagement, Non-musicians, Music and Color.

I. INTRODUCTION

In the context of digitization, the need of creating specific digital music has expanded beyond the recording industry thanks to accessible, software-based musical instruments. Young generation outside of the professional community are using this technology in personal projects or music education. Indeed, music and sound effects can be combined with any digital or tangible creation to produce audiovisuals, such as YouTube videos, films, advertisements, theatrical performances, videogames and so on. Among them, content creation is becoming a trend among young people.

According to HubSpot's 2022 State of Consumer Trends study, 30% of those surveyed between the ages of 18 and 24 and 40% of those between the ages of 25 and 34 identify as content creators[1]. The use of music in content creation is omnipresent. Digital Audio Workstations (DAW) are used to cover all stages of the music creation process[2] and can interest content creators or anyone that may not have a background in music theory. Unfortunately, the content creator's ability to use DAWs depends on his/her background in digital music[3], so these amateurs may feel lost in front of the music software. Plus, they do not necessarily want to waste their time and energy learning another domain such as

audio engineering or music theory. Therefore, content creators go through alternatives to music composition. Most of the time, they use music that is directly available on the content sharing platform, for instance TikTok “has licensing agreements with those artists allowing them to offer that music to the end-user for use in TikTok videos”[4]. For creators that publish videos on platform that do not directly offer music to video editors such as YouTube, they take advantage of royalty-free music like Epidemic Sound[5]. Otherwise, they can use copyrighted music in their creation, but they must have the permission of the owner of the rights.

To make audiovisuals closer to the creator's expectations, audio production should be done ideally by the same person or entity just like he/she/it does for the visual part. Between the software intended for professionals and the alternatives cited below that do not entirely make him/her/it the author of the work, there is not yet a clear solution that has been developed to permit any content creator to produce music to support their creation without requiring a musical background. How can a DAW be developed for non-musicians? The main idea developed in this project is to analyse the complexity of music production softwares in order to decrease it.

The complexity decrease process involves supporting non-musician's creative engagement by designing a new music production software. The interface is designed to embody a sequencer that can be integrated in an existing open-source DAW using key findings in the fields of New Interfaces for Musical Expression (NIME)[6], Relationship Between Music and Emotions and Creative Engagement. The software focuses on the music production in a specific mode of musical performance that speaks to content creators: Muzak, alias background music.

The research question can be described as: How a music production software design can guide and support non musicians' creative engagement?

II. SCIENTIFIC CONTEXT

A. Musicking

In the NIME field the term ‘musicking’ describes a more accessible music making activity that is not exclusive to trained musicians[7]. This trend has produced Interactive Music Systems designs (IMS) which have enabled non-musicians to actively play with music rather than only passively listening to music[8]–[10]. In this project, non-musicians are considered as amateurs of musicking. They are interested in musicking activities but with no desire to become professional musicians, as non-musicians have less access to formal music training. It is said to be difficult for non-musicians to acquire musical inventiveness, which has frequently been thought to the exclusive domain of professionals [11]. According to studies, non-musicians find it challenging to create original musical ideas because they lack the conceptual and technical knowledge and skills necessary to create music. (ibid).

The above literature defines the terms ‘musicking’ and ‘non-musician’ that represent the activity and the user. The definitions are essential to delimit the research question and therefore the project. The project will not address anyone with a musical background or professionals in the record industry.

B. Digital Audio Workstation

In the opposite of IMS aimed to musicking, Digital Audio Workstation (DAW) is a music production software used to produce music for the record industry (Fig. 1). The first DAWs were developed in the late 1970s and early 1980s, in parallel with the advent of personal computers[12]. DAW can be described as a collection of virtual tools that aims to produce music. It is a setting where sound design and sample editing tools coexist side by side with loop-trigger and sequencer paradigms[13]. A loop trigger refers to a mechanism that initiates the playback of a pre-recorded musical loop while a music sequencer is a device or application software that can record, edit, or play back music. Users can produce music by either recording live session through a MIDI controller that usually represents a piano (if they can play an instrument) or interacting with the sequencer displayed on the interface. It also includes virtual recreations of conventional musical instruments and notation systems.



Fig. 2. Interface of a common DAW, Logic Pro. The aspect of a multi-track tape recorder can be seen in the header. The loop-trigger and sequencer paradigm can be exploited on the center, each track representing an instrument, while the MIDI editing appears on the lower part.

The user interfaces of many of the most widely used DAWs are de facto difficult to comprehend for beginners since they are intended for professionals[2]. Such complexity is partly due to their skeuomorphic design (Fig. 2). Skeuomorph design incorporates real world qualities of goods into the digital GUI[14] in order to create a familiar feeling and raise the perceived worth of products[15]. If a person is familiar with a real-world object, they may interact with its digital metaphor more quickly and naturally. However, tools like analog mixing consoles, MIDI controllers or analog synthesizers are not familiar to non-musicians. In fact, the use of hardware metaphors such as analog mixing consoles[16] and vintage analog synthesizers[17] is omnipresent.



Fig. 2. Emulation of the first totally programmable synthesizer on the market with digital vector synthesis (on the right).

Skeuomorphs frequently follow generations, making them time sensitive. The original meaning of a skeuomorph, which was deeply ingrained in one generation, might not be relevant or useful to the subsequent generation. Studies show a significant difference in terms of emotion experiences, preferences, visual search, identifying clickable objects, and multiple page navigation, between generations depending on

the type of interface[14], [18], [19]. In other words, skeuomorphic UI design is more adapted to elder people and Flat design speak to young generation and young adults.

The DAW's abundance of add-ons, such as Virtual Studio Technology (VST) plugins[20], with new ones being added on a regular basis, contributes significantly to its power and therefore its complexity (Fig. 3). VST is materialized by a software interface that integrates synthesizers (electronic instrument) and effects units into DAWs. VST and similar technologies use digital signal processing to simulate analog recording studio hardware into software. Thus, sample packs increase the size of audio file libraries, VST plugins increase the DAW's flexibility, and music software is being improved and complexified.



Fig. 3. Interface of Kontakt, a software sampler that integrates VST. This is a drum set that has been virtualized to be playable for users.

In fact, attempts to decomplexify the UX/UI has been made with mixed results. GarageBand[21] stands out among other DAWs for its simplicity of use (Fig. 4), but some improvements are to be made according to Bell's affordance evaluation with the principles of presumptions, privileges, provisions, protections, and preventions[16]. Groovy Music(Fig. 5) is a model DAW that compromises depth for simplicity[22]. Right away, the childlike design of the product was particularly appealing to a younger audience while minimizing the amount of functionality lost. It succeeds in imparting the fundamentals of using DAWs in a simple-to-understand manner.



Fig. 4: Interface of GarageBand

Sugar Pucks(Fig. 6) is a combination of video game and music software[23]. Like Groovy Music, the interface is simple and appealing. The use of color as indicator and the infantile dimension are also dominant. In turn, the functionality awarded to the user is more limited than the average DAW.



Fig. 5 - 6: Groovy Music (top) and Sugar Pucks (bottom)

These UI designs cannot be called as skeuomorphic, but more like a childish design. These DAWs still can serve as examples of how visual appeal may be used to emphasize the simplicity of our designed DAW, while putting aside their childish aspect and increasing the number of functionalities. Finally, it is good to mention that an attempt has been made to create a DAW using audio-haptic interfaces[24].

The above literature explains what a DAW is, its features and its complexity related to the interface. It is showed how difficult is for non-musician to use DAWs for personal

projects. The project described here aims to find a solution so that non-musicians can have an environment where they can be able to create their own music.

C. Creative Engagement

Three levels of engagement are identified for an interactive systems user: sustained engagement, immediate engagement, and creative engagement[25]. The latter emerges from creative activities when a user is engaged in a reflective, constructive, and active cognitive process in making a creative outcome with an interactive system[26]. Since music making is a key form of creativity[27], a design of interactive support for creative engagement can be explored in the domain of DAWs.

Empirical study shows that in an IMS, creative engagement can be supported with features such as ‘learn the sound’, ‘playing live’, ‘catalyze insights’ and ‘provide a starting base’ [28]. Since IMSs refer to systems that produce music from the input of non-expert users[29], these findings are applicable for the designing of a DAW intended for non-musician .

In the music composition process, DAWs can be used for more than arranging and producing the song in accordance with an established model. When composing through a software, musical ideas generally come from generating unexpected novel ideas and are supported by experimentation in the software and the need for visual representation[30]. In fact, the software can be considered as an instrument for singer-songwriters and permits to create musical ideas by exploring its tools and effects[31][13]. Such findings conclude to say that there is no need to be a musician to create musical ideas, as long as the person is actively engaged in the music production software.

The above literature supports the possibility of designing an interactive support for creative engagement within a DAW.

D. Visual Stimuli

Stimuli from visual materials are essential for facilitating creativity[32]. They can positively prompt creative performance since they provide ‘potential cues, analogy-sources or other similes’ for inspiration[33]. More specifically, abstract visuals are effective external stimuli to guide novices to find inspiration and to support creative engagement[34]. For example, painting and music are two fields that can be associated[35] to form a painterly interface that use expressive gestural drawing as an input to control the sound output[28], [36], [37]. In music making, graphical scores (GS) have been proposed as an open tool in visual form (Fig. 7) for both trained and unexperienced musicians to create music[38]. A GS is a collection of personal visual representations that transmit different facets of the audio information necessary to perform a piece of music.[39]. In addition, GSs encourage performers to interpret and decide what to play in their own way, which allow performers to consider themselves as composers[40].

Novices tend to follow the examples of musical ideas (e.g. combinations of samples and rhythmic patterns) suggested in the visual stimuli and reproduce them[34].

Findings indicated below support the idea that non-musicians can create their own GS and have a starting base in a context of supporting their creative engagement in musicking.

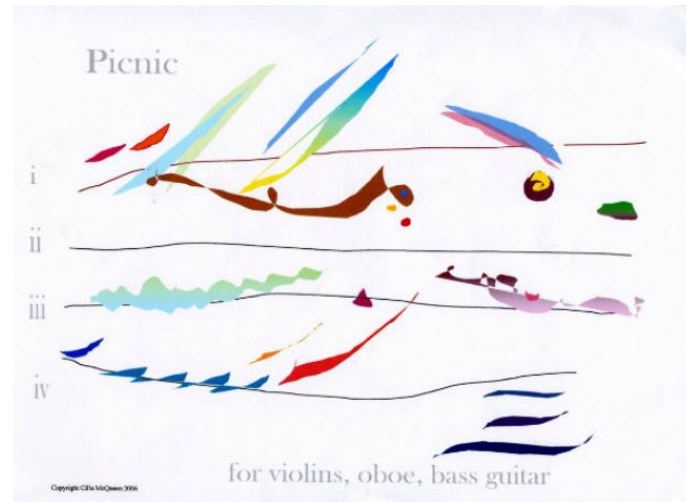


Fig. 7. Graphic scores that represent the music piece ‘Picnic’ by McQueen. “Each line represents a different instrument, with the colours and shapes informing how the music might sound.”[41]

E. Artificial Intelligence tools

Artificial Intelligence (AI) has made significant advancements in the field of music composition, revolutionizing the way we create and appreciate music. AI systems can now generate original compositions, assist musicians in the creative process, and even compose music in collaboration with human artists [42].

AI systems that compose music typically utilize machine learning techniques, particularly neural networks[43]. AI algorithms generate music by combining different musical elements such as melody, harmony, rhythm, and texture. These algorithms use statistical models, probabilistic methods, and rule-based systems to create coherent musical compositions[44]. They can also incorporate variations, improvisation, and randomization to make the generated music more diverse and unpredictable[45].

AI algorithms can be trained to compose music that elicits specific emotions or moods[46]. By analyzing emotional cues present in music and leveraging techniques such as sentiment analysis, AI systems can generate compositions tailored to evoke joy, sadness, excitement, or other emotional responses[47]. This capability opens up new avenues for using music as a therapeutic tool or enhancing multimedia experiences.

Rather than replacing human musicians, AI is often used as a tool for collaboration. Musicians and composers can use AI systems to assist them in the creative process by generating musical ideas and providing inspiration. Dozens of generators exist for chord progressions such as ChordChord[48] or ChordCreate[49], and also for melodies such as MelodyStudio[50] or DopeLoopAI[51]. This collaboration between human creativity and AI algorithms can lead to new and exciting musical expressions.

The scientific advancements in AI have significantly impacted the music industry, creating new opportunities for artists, composers, and music enthusiasts alike. For non-musicians, there is still a learning process before knowing the existence of such AI-based technologies. To search for a chord progression generator, they must know what a musical chord is. In addition, generators either bring a starting base or give a complete song without any human interventions, leading to an open question about the place of human creativity in AI-based music. Our project aims to give non-musicians musical ideas while supporting their creativity engagement to create a song that they can claim to be the creator of.

III. PROPOSITION

The proposition builds upon the research findings and theoretical frameworks mentioned earlier. It aims to simplify the music creation process by guiding the composer through the music making and by supporting the user's creative engagement. To do so, a new music production software design based on the experimental results and theories described above has been developed. It is a combination of two generators and a music sequencer that creates a GS.

A. Chord Progression Generator

The first component of the software design is the chord progression generator module, which utilizes a graph traversal algorithm to generate musical ideas based on user input and preferences. This module takes into account the user's desired genre and mood to generate harmonic and rhythmic patterns that represent the chord progression of the song.

As far as the association between emotion and chords is concerned, there is no scientific consensus in the field of Music and Emotion. To build our algorithm, we chose Berkemeier's Chord Relationship and Emotion[52] since it focuses on basic chords for novices. The theory associates several chords suite to an emotion. Each chord relationship has a detailed description of the mood (Fig. 8).

Spectrum	Emotion	Chord Relationships
Sad	Sadness	M III m
	Tragic	m IV m / m V m
	Bitter Sweet	m V M / m vii M / M II m
Evil Sounding	Mysterious and Tense	m ii m / m VII m / m VI M / M iii m
	A Little Uneasy	m II m / m vii m
	Otherworldly and Ominous	m iii m / m VI m
	Ominous and Dark Sounding	m III m / m vi m
	Antagonism and Danger	m IV+ m
Evil/Neutral Sounding	Dramatic	m VII M / M ii m
	Mysterious/Dark comedy	m II M / M vii m
	Outer Space	m IV+ M / M IV+ m
Neutral Sounding	Exotic/Cowboy/Enchanted Forest	M ii m / M VII M
	Outer Space	M IV+ M
	Cautious but Optimistic	M VII m / m ii M
	Rising Action/Tension	m iii M
	Powerful and Mysterious	m III M / M vi m
	Resolution	m vi M
Good/Neutral Sounding	Romantic/Exotic	M IV M / M V m
	Other worldly/Heavenly	M VI m
	Wonder and Transcendence	m IV M / M V m
Good Sounding	Protagonism	M II M / M vii M
	Heroic	M iii M / M VI M
	Fantastical	M III M / M vi M
	Good Energy	M IV M / M V M

Fig. 8. Berkemeier's Chord Relationships and Emotion.

With Berkemeier's theory and some notions of solfège, a graph can be created where each node represents a chord, and each arc represents an emotion (Fig. 9). The whole graph contains all the possible chords in a specific musical scale and therefore a chord progression in a particular mood can be constructed by navigating through the path.

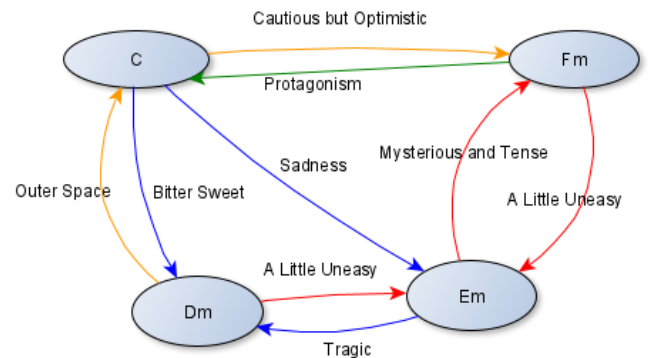


Fig. 9. Example of graph with 4 chords and 9 relationships in C major. To simplify the lecture, some arcs and nodes have been removed.

The component also provides real-time feedback as the user chooses the parameters. By providing these pre-composed musical elements, the software assists non-musicians in overcoming the initial hurdles of creating music from scratch,

allowing them to focus on their creative expression rather than technical intricacies. In addition, a noticeable hypothesis for the prototype design can be raised. The chord-emotion relationship will be useful to determine whether the link between emotion and color is applicable to musical chords.

B. Melody Generator

The second component of the software design is another generator module, which generates the melody that can be repeated throughout the music. This module employs a combination of algorithmic composition techniques and sample libraries to produce a variety of instrumental and electronic sounds.

The music theory has been put in the algorithm to avoid false notes and dissonance. Like the previous component, the generator is instantly played so that the user understands the outputs of the generator. The generated melody adapts to the first generator, providing a harmonic and rhythmic body that enhances the overall musicality of the piece.

C. Graphical Scores and Sequencer

To visualize the composition and facilitate user interaction, an interactive GS is implemented within the software. It can be manipulated through a music sequencer, which enables users to arrange and structure their compositions. The GS is timeline-based and users can place and manipulate the musical elements generated by the software. It allows intuitive editing, arranging, and looping of different sections, making it accessible and engaging for non-musicians who may not be familiar with traditional musical notation. Multiple prototypes have been developed (Fig. 10) and are intended to be selected by potential users.

Each GS features a minimalistic and clean design, embracing the principles of flat design. Uncluttered and devoid of unnecessary embellishments, the interface focuses on presenting the musical elements in a visually appealing and straightforward manner. The graphical scores utilize bold colors and simple geometric shapes, in line with the flat design philosophy. The visual representations reproduce the main characteristics of a note, i.e. the pitch, the duration and the intensity.

With a flat design approach, the graphical scores prioritize intuitive and accessible visuals. Users can easily comprehend the musical structure and arrangement by observing the graphical representation, promoting an effortless understanding of the composition.

The flat design of the graphical scores extends to the interactive elements, emphasizing user-friendly interactions. Users can seamlessly manipulate and rearrange musical elements by employing simple drag-and-drop actions, enabling a smooth and intuitive composition process.

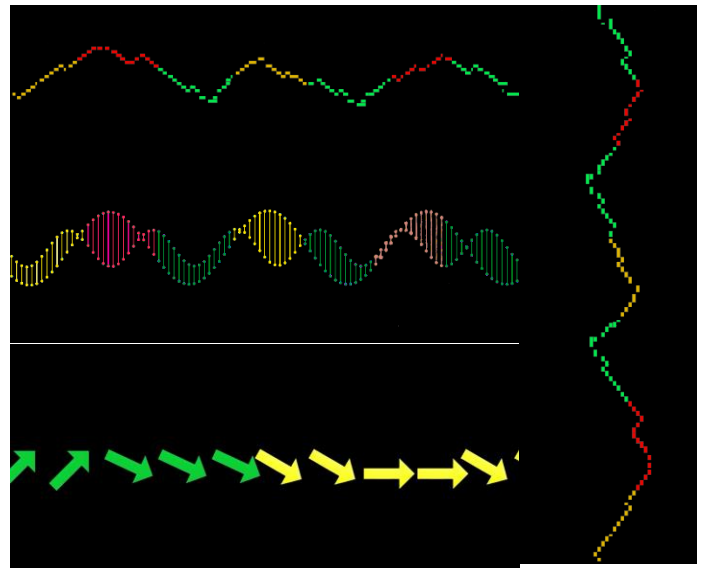


Fig. 10. We created 4 different GS that represent the same song using flat design.

Overall, the proposed music production software design aims to bridge the gap between non-musicians and music creation by guiding and supporting their creative engagement. By leveraging machine learning algorithms, algorithmic composition techniques, and an intuitive GS interface, the software empowers non-musicians to express their musical ideas, explore different genres, and create compositions that reflect their unique artistic vision. This flat design not only simplifies the music creation process but also fosters a sense of creativity and musical exploration among non-musicians, expanding the boundaries of music making and enabling a broader participation in the art of music composition.

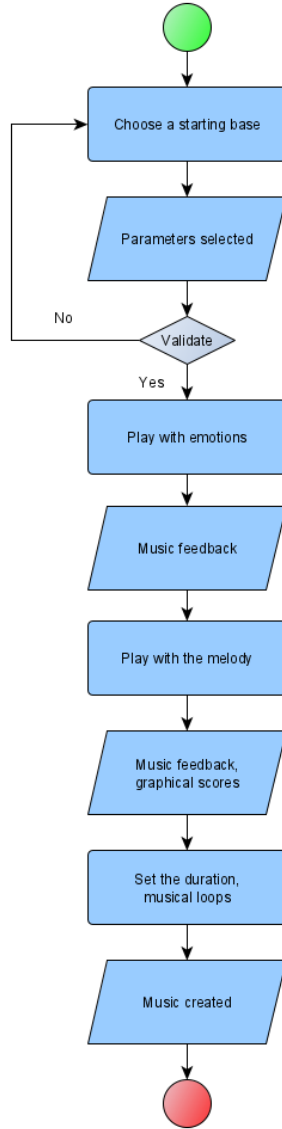


Fig. 11. Flowchart of the proposition with the 3 components: the chord progression generator, the melody generator and the sequencer that integrates a graphical score.

IV. PROTOTYPE DESIGN

The instantiation of the proposition involves providing a user interface (UI) for each component. The UI design based on an online survey with 27 participants. The aim of this survey is to determine the best suitable UI and to explore the potential transitivity between chords and colors, leveraging the existing associations between chords and emotions as well as emotions and colors.

A. Analysis of user preferences

To begin, an online survey was conducted with a diverse group of participants, including both musicians and non-musicians. The survey consists of a questionnaire that aims to identify consistent patterns and preferences in the chord-color

associations, providing a foundation for the user interface design.

1) Colors, Emotions and Music

The Plutchik's Wheel of Emotions (Fig.12) is often described as a reference in associating colors and emotions[53] and therefore chosen to determine the chord-color associations. In the survey, the participants were presented with an audio file in which various chords were played using a piano. They were asked to listen to the audio and associate the chord with colors based on the emotions they evoked. The chords were those initially associated with an emotion according to Berkemeier's theory. The question was whether the selected color matched the Plutchik's color of the Berkemeier's emotion of the chord.

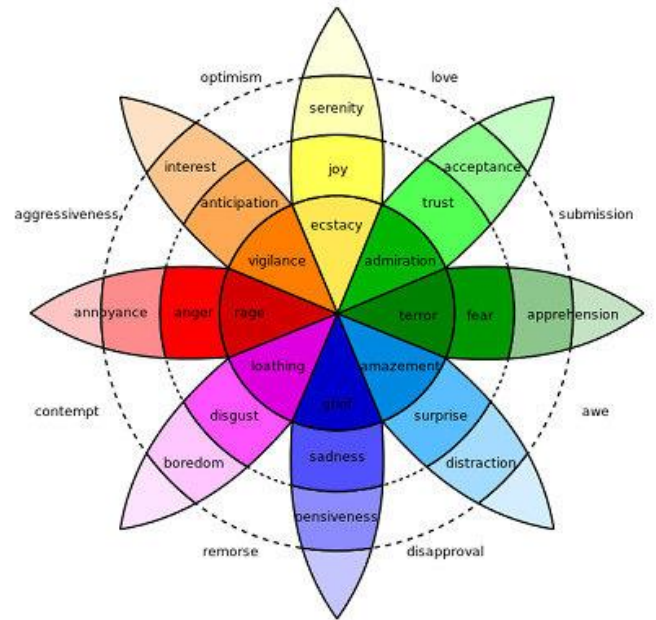


Fig. 12. The Plutchik's Wheel of Emotions[54].

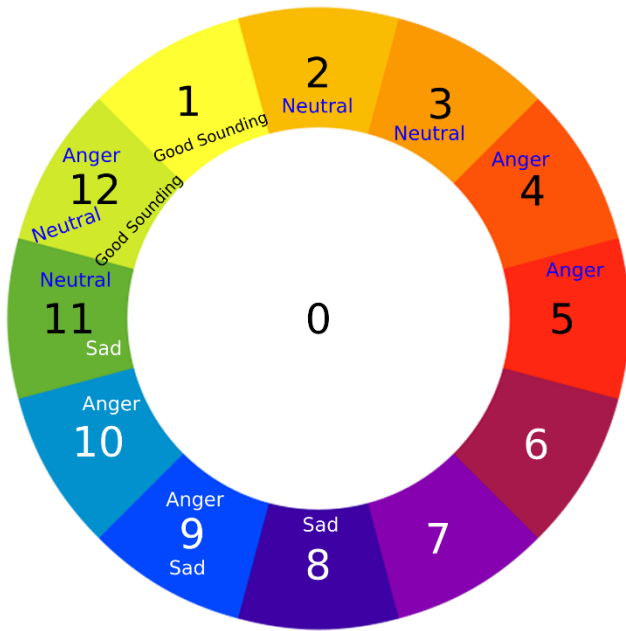


Fig. 13. The results of the study coincide nearly with the Plutchik's Wheel of Emotions. The emotions in this wheel represent those associated with the chords that participants have listened to.

The results of the survey revealed a range of chord-color associations, indicating a certain level of subjectivity and personal interpretation (Fig. 13). However, a significant number of participants demonstrated consistent preferences in their chord-color associations, suggesting a potential for transitivity between chords and colors specially for “Sadness” or “Anger”.

Based on these findings, the prototype UI was developed to reflect the chord-color associations observed in the survey results (Fig. 14). A new color wheel is created with the combination of Plutchik's wheel and the outcomes of the survey. By associating colors with chords and leveraging the transitivity between chords, emotions, and colors, the interface aims to bridge the gap between non-musicians and the complexities of music theory and composition.



Fig. 14. The colors chosen for the emotions on the interface.

The generators and sequencer within the prototype provide users with the tools to manipulate and arrange the generated musical elements. The color associations derived from the survey are then used to map these musical elements to specific colors.

2) Graphical score testing

Different variations of GS mentioned earlier were also presented to the participants. They had to evaluate the pertinence of each image in a scale of 0 to 5. The chord that accompanies the melody is represented by one of Plutchik's color. A modified version of the actual music representation in several DAWs was also displayed to the participants.

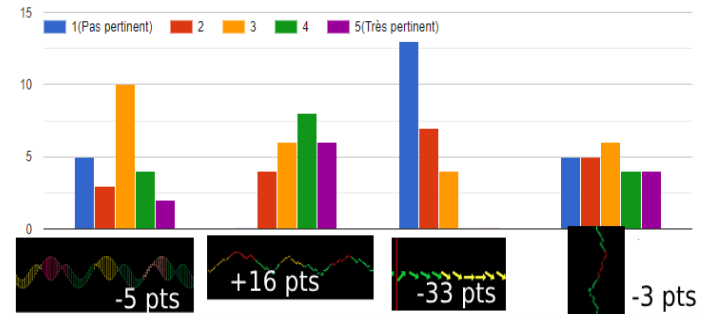


Fig. 15. The results of the study for the most accurate GS

The results (Fig. 15) showed a significant preference for one representation that can be used for the sequencer. Within the sequencer, users can arrange the generated musical elements on a timeline-based interface, with each color representing a specific chord or musical phrase. Users can easily manipulate the composition by dragging and dropping the colored elements to create different sections, variations, and repetitions.

B. Prototype interface

The prototype encourages non-musicians to engage creatively with the music-making process by providing a visually intuitive and accessible user interface. It is designed with a modern, flat design aesthetic, emphasizing simplicity, minimalism, and clean visual elements. It draws inspiration from video games, incorporating familiar elements such as a play feature, game menu, and settings. The interface aims to provide an intuitive and engaging user experience for both musicians and non-musicians.

1) Home Screen

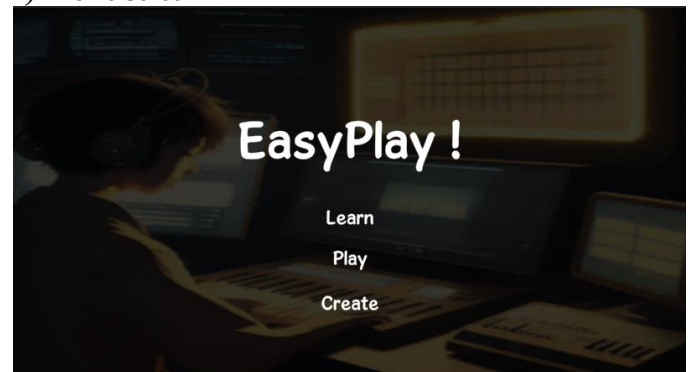


Fig. 16. Home Screen of the prototype

Upon launching the software, users are greeted with a home screen that serves as the central hub for accessing various features and functionalities (Fig. 16). The home screen is designed with a minimalist layout, presenting the user with a clean and organized interface. The Learn feature is a tutorial that explains the program step by step. The Play and the Create features lead to the sequencer. The difference is that the Play feature is where the generators feature are being manipulated while the Create feature starts the graphical score from scratch without any generated content.

2) Parameters of the generators

The generator parameters menu layout is based on Spotify's music selection menu with a clean menu (Fig. 17). The user is invited to select a genre among others such as Bossa Nova, Hip-hop, Jazz, Dance, Rock or Lo-fi. There are in fact the most used genre on TikTok[55]. Each time the user changes the genre, an audio preview of a rhythm is played. Then the user must choose an emotion. The available emotions are Happiness, Sadness, Neutral (provides a neutral sounding) and Evil (provides an evil sounding). The parameters selection section ends up with the selection of two instruments (one for the bassline and one for the melody). The available instruments are piano, violin, synthesizer, guitar, saxophone, and electric guitar. These rhythms and instruments come from sample libraries of several open-source VST plugins.

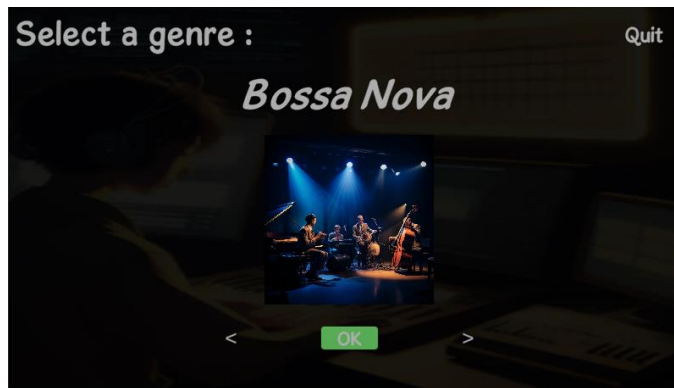


Fig. 17. Generator parameters selection screen.

Once the user has selected all the parameters, a chord progression and a melody are generated from the graph traversal and the music theory algorithms. A chord progression consists of 4 chords, so 3 chord relationships (one relationship between 2 chords). For the user, the chord progression consists of a list of emotions displayed in the header. The emotion description of each chord is colored according to the color wheel. The user has the possibility to either remove or add more chords. In the middle of the screen, the color wheel displays the available chords depending on the previous chord (Fig. 18). In the lateral section, the user can select an emotion. Each time the user presses an emotion, the corresponding chord is played, providing users with real-time audio feedback of their musical creation. When the user is satisfied with the list of emotions, he/she can reach the next

page that contains the graphical score and the sequencer interface.

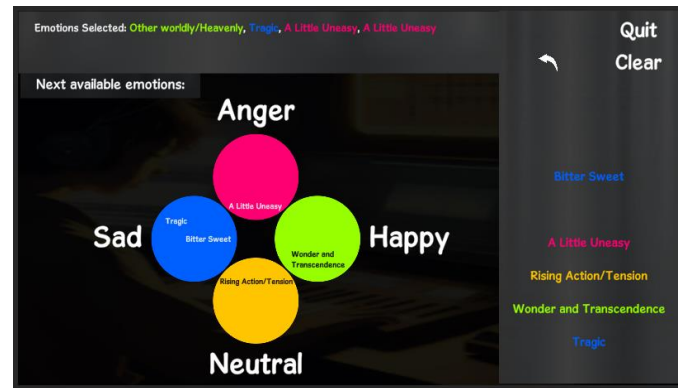


Fig. 18. The colors chosen for the emotions on the interface.

3) The sequencer interface

The sequencer interface is a central component of the music production software (Fig. 19). It provides a timeline-based representation of the composition, enabling users to visually organize and manipulate the generated musical elements. Users can drag and drop the generated musical patterns onto the timeline. Each musical element is represented by its associated color, creating a visually appealing graphical score that facilitates intuitive manipulation and arrangement of the composition.

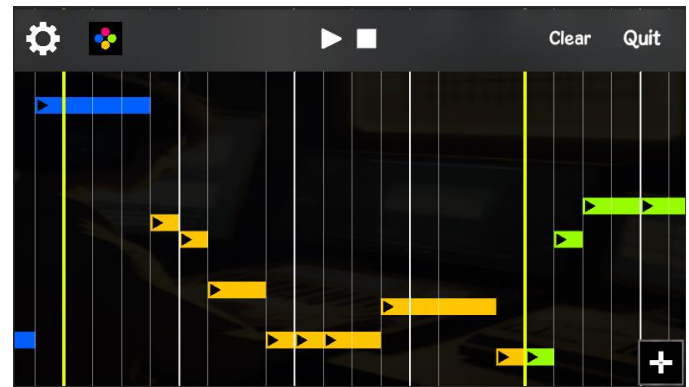


Fig. 19. Sequencer interface with the interactive graphical score in the center.

a) Timeline View

The sequencer interface features a horizontal timeline view that spans the duration of the composition. The timeline is divided into measures or beats, depending on the desired level of precision. Each measure represents a fixed unit of time in the composition, allowing users to align and position musical elements accurately.

b) Color-Coded Musical Elements

To align with the color associations derived from the chord-color survey, the musical elements within the sequencer interface are color-coded. Each musical element, such as a melody or chord progression, is represented by a specific color that corresponds to the associated emotion or mood. This

visual representation allows users to identify and differentiate different musical elements briefly.

c) Drag-and-Drop Functionality

The sequencer interface supports intuitive drag-and-drop functionality, enabling users to place and move musical elements within the timeline. Users can select a musical element and effortlessly drag it to a specific position on the timeline to determine its starting point and duration. This flexibility empowers users to experiment with different arrangements and easily create variations within their composition.



Fig. 20. Sequencer interface settings.

d) Looping and Repetition

The sequencer interface supports looping and repetition features, allowing users to create repeating sections within their compositions. Users can select a section of the timeline and define it as a loop, which repeats continuously during playback. This functionality is particularly useful for creating catchy hooks, building tension, or developing rhythmic patterns.

e) Melody Patterns

In addition to the previously described features, the sequencer incorporates a unique and innovative element—an integrated melody pattern display (Fig. 21). This feature offers users a collection of pre-designed graphical patterns that can be utilized to create melodies effortlessly.

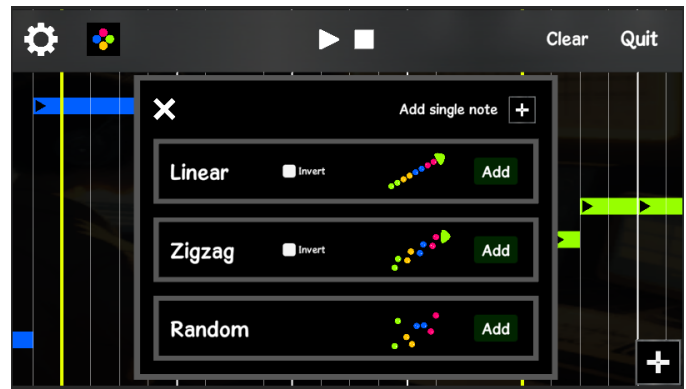


Fig. 21. Melody patterns menu.

Overall, the interface of the music production software combines a flat design aesthetic with elements inspired by video games. It incorporates a play feature, game menu, and settings section to enhance usability and engagement. The visual representation of generated musical elements through color associations adds a unique and visually stimulating aspect to the interface, providing users with an intuitive and creative environment for music composition.

Through user testing and feedback sessions, the prototype's effectiveness in guiding and supporting non-musicians' creative engagement will be evaluated. The feedback received will help identify any potential improvements and refinements to the prototype design, ensuring a user-friendly and empowering experience. The upcoming user testing phase will play a crucial role in assessing the effectiveness of the prototype in facilitating non-musicians' creative expression and engagement in the music-making process.

V. EVALUATION

The evaluation of the proposed solution consists of assessing whether the software effectively supports non-musicians' engagement in creative music composition while providing guidance throughout the process. The primary objective of this evaluation is to determine the software's ability to empower individuals without formal musical training to express their creativity and create meaningful musical compositions.

A. Test procedure

A comprehensive comparison was conducted among three groups of participants: 9 non-musicians utilizing the software, 8 non-musicians using a basic digital audio workstation (DAW), and 3 local experienced artists. Non-musicians are invited to compose a song in a mode of background music. The evaluation focused on examining the creative outcomes, the creative potential of the software as perceived by non-musicians, and overall quality of the compositions, with specific consideration given to their repetitive and simple

nature, which make them suitable for accompanying other media such as videogames or YouTube videos.

To ensure anonymity and unbiased results, the entire test and evaluation process was conducted anonymously. Non-musician participants were not required to disclose their personal information, and their identities remained confidential throughout the evaluation. This approach aimed to create a comfortable and non-judgmental environment for participants, allowing them to freely explore their creative potential without concerns about their personal identity being associated with their compositions.

1) EasyPlay! testing

The test process involved inviting non-musician participants to create songs using the software, following a five minute tutorial that familiarized them with its features and functionality. We let them twenty minutes to create a song. After the creation process, semi-structured interviews were conducted to gather valuable feedback from these participants.

The seven minutes interviews aimed to gain insights into their perceptions of the software's usability, its ability to facilitate their creative engagement, and whether it effectively guided them through the music creation process despite their lack of formal musical training. Questions about their perception of the software such as "Do you consider this software to be creative?" or "Did you feel a sense of control and impact over the music you created using the software? If so, how?" were asked.

2) Reaper testing

To provide a comparative assessment, songs created by non-musicians using a basic DAW called Reaper[56] were also included in the evaluation (Fig. 22). This allowed for a direct comparison between the output of the EasyPlay! software and a more traditional music production tool.



Fig. 22. Interface of the sequencer (Reaper).

The procedure to follow is the same as the first group, including tutorial, song creation task and interview. The

sample libraries are similar in the quality, with the difference that each note can be played in a MIDI controller rather than in the sequencer only and no starting base was generated.



Fig. 23. Interface of the instrument settings (Reaper).

3) Creativity Evaluation

Additionally, songs composed by artists served as a benchmark to gauge the level of creativity and quality achieved by experienced musicians. The artists were chosen for their specialty, which is the creation of background music. They were simply asked to give permission to be evaluated on their previous song.

One person studying musicology in the University of Geneva, another student in the Haute Ecole de Musique and a music expert were also involved in the evaluation. Their role was to provide professional assessments and comparisons of the compositions created by both non-musicians and experienced artists. The expert and students evaluated factors such as musicality, arrangement, simplicity, and suitability for multimedia applications. It is worth noting that they were not specifically asked to assess the sound and audio quality of the music created using the software. This distinction is due to the fact that the songs produced through the software utilize the same open-source sample library, ensuring consistency in the audio quality across all compositions. The insights of experts and music students contributed to the overall assessment of the software's performance and its impact on non-musicians' creative engagement.

B. Qualitative and Quantitative Methodologies

The interviews conducted as part of the evaluation were semi-structured, which aligns with a qualitative evaluation approach. This approach allowed for in-depth exploration and understanding of the participants' experiences using the software, their creative engagement, and their perceptions of its usability and effectiveness. The qualitative data obtained from these interviews was subsequently analyzed using the thematic analysis methodology to identify recurring themes,

patterns, and insights within the participants' responses.

To gather quantitative data, a questionnaire was designed specifically for this purpose by the expert, utilizing a methodology commonly employed in the field of music for evaluating students' compositions. The questionnaire consisted of structured items and rating scales to systematically capture the evaluator's assessments of various aspects related to the compositions created by the composer. The questionnaire might have included items related to musicality, arrangement, repetitiveness, suitability for multimedia applications, or any other relevant dimensions of evaluation.

The quantitative data collected through the questionnaire allowed for a more objective and measurable assessment of the compositions, providing numerical values that could be analyzed statistically. This approach facilitated comparisons and generalizations about the quality, effectiveness, and overall impact of the software on non-musicians' creative engagement.

By combining both qualitative and quantitative evaluation methods, the test aimed to obtain a comprehensive understanding of the software's performance and its impact on non-musicians. The qualitative analysis of the interviews provided rich insights into participants' experiences, perceptions, and subjective aspects of their creative engagement, while the quantitative assessment through the questionnaire offered quantifiable data to support broader evaluations and comparisons.

This multi-method approach ensures a well-rounded evaluation, incorporating both qualitative and quantitative perspectives, and provides a comprehensive view of the software's effectiveness in supporting non-musicians' creative engagement while guiding them through the music creation process.

C. Data Analysis

The evaluation of the songs created using our music production software involved a combination of qualitative and quantitative data collection methods. The qualitative data was obtained through semi-structured interviews conducted with non-musician participants, while the quantitative data was gathered through a questionnaire completed by music experts and musician students.

1) Qualitative Data Analysis

Thematic analysis was employed to analyze the qualitative data obtained from the interviews with the non-musician participants. The interviews were recorded and transcribed, and key themes and patterns were identified within the participants' responses regarding both softwares. These themes included their experiences with the software, their perceptions of its guidance and support in the music creation process, and

their overall feedback on the user interface and creative engagement. The analysis of the qualitative data showed several important insights.

a) *Guidance and Vibe*

Participants expressed more positive experiences with the first software than the DAW, highlighting its user-friendly interface and the guidance it provided throughout the music creation process: "I like the fact that it can give me a framework and then I can change according to my desires". This theme emphasizes the software's ability to assist non-musicians in achieving the intended emotional atmosphere in their music. They appreciated the graphical nature of the melody patterns and found them helpful in creating melodies even without prior musical knowledge.

For this Reaper, three participants said that the tutorial was not long enough to fully understand the place of each feature. The song created by most participants remained unfinished, giving them a feeling of frustration.

b) *Enjoyment and Exploration*

All the participants expressed a desire for more time to explore the various alternatives and features offered by the software. This theme suggests that participants found the music creation process enjoyable and engaging, indicating a positive user experience: "If you'd given me more time, I'd have had fun testing all the alternatives". On the other hand, only two participants mentioned their appreciation of the exploration phase of the DAW interface.

c) *Impact and Control*

Participants appreciated the ability to choose instruments and make changes to the composition based on their satisfaction. This theme highlights the sense of impact and control that participants felt, allowing them to shape the music according to their preferences and creative vision. The DAW features the same control and participants also mentioned it through the interview.

d) *Feedback and Iteration*

Three participants mentioned the desire to listen to the selected emotion again from the beginning, suggesting the importance of providing comprehensive feedback and the ability to review and refine compositions. As mentioned before, the corresponding chord is played when the user has selected an emotion, providing users with real-time audio feedback of their musical creation. However, the chord progression's feedback feature, that is playing one chord at a time, is available only once the user has selected enough number of emotions. This theme underscores the value of iterative processes and the opportunity for participants to make informed decisions during the music creation journey.

Since Reaper is a more sophisticated software, its functionalities were well mastered according to the

participants, who therefore appreciated the feedback function.

e) *Perceived Creativity*

It was found that participants using the first software perceived it as a valuable tool for enhancing their creativity. They likened the experience to “playing with Lego”, suggesting that the software provided them with a platform for exploration and experimentation in their music creation process.

In contrast, participants who used the basic DAW expressed difficulties in composing music effectively. They reported challenges in creating harmonious and pleasing compositions, often resulting in dissonance. This suggests that the DAW might have posed greater complexity or barriers for non-musicians, making it harder for them to achieve the desired musical outcomes.

By analyzing these responses, it becomes evident that the themes of guidance, enjoyment, impact, feedback, and perceived creativity stand out. These themes reflect the participants' experiences, perceptions, and feedback regarding the software's effectiveness in supporting their creative engagement and providing a user-friendly environment for music creation.

It is important to note that these results are based on participants' subjective perceptions and experiences. The quantitative analysis could involve assessing the actual musical output of the participants to provide a more comprehensive understanding of the creative outcomes and the potential impact of the software on non-musicians' compositions.

2) *Quantitative Data Analysis*

The quantitative data obtained from the questionnaire responses of music experts and musician students was analyzed using statistical analysis techniques. The questionnaire consisted of structured items and rating scales to assess various aspects of the compositions created using the software.

The data analysis revealed average ratings for musicality, arrangement, repetitiveness, suitability for multimedia applications, creativity, and overall quality of the compositions. Visual representations, such as bar charts and graphs, were used to present the quantitative findings. These findings provided a numerical basis for comparing the compositions created by non-musicians using the software with those created using a basic digital audio workstation and those composed by artists.

The descriptive statistics were calculated to summarize the ratings provided by the participants for each category, including musicality, arrangement, repetitiveness, suitability for multimedia applications, and creativity. The mean ratings were computed to provide an overall measure of the

participants' perceptions.

a) *Hypothesis*

The songs created with the software have received a higher average rating compared to the songs created with the DAW and the songs created by artists.

The null hypothesis (H0) states that there is no significant difference in the average ratings among the three categories. The alternative hypothesis (H1) proposes that there is a significant difference, with the songs created using the software receiving a higher average rating.

b) *Data Analysis*

To test this hypothesis, a statistical analysis was conducted using Analysis of Variance (ANOVA). The ratings provided by the experts were used for comparison.

The significance level (alpha) was set at 0.05, indicating a 5% threshold for statistical significance. If the obtained p-value is less than 0.05, we will reject the null hypothesis and conclude that there is a statistically significant difference in the average ratings among the categories.

The analysis involved calculating the sample means and sample standard deviations for the ratings of songs created with the software, songs created with the DAW, and songs created by artists. An ANOVA test was performed to determine if there are significant differences in the means across the groups (Table 1).

Origin of the songs	Mean (0 to 12)	Standard Deviation
Songs created with EasyPlay!	7.82	1.21
Songs created with Reaper	7.45	1.34
Songs created by artists	8.15	1.10

Table 1. Mean and Standard Deviation of the songs created in each category.

The ANOVA test yielded the following results:

F-value: 3.42

p-value: 0.042

The F-value represents the ratio of between-group variability to within-group variability. The p-value indicates the statistical significance of the observed differences.

Based on the obtained results, the ANOVA test revealed a statistically significant difference in ratings among the three categories of songs ($p < 0.05$).

D. ANOVA Results

In the quantitative analysis, the results of the ANOVA analysis indicate that there is a significant difference in the average ratings among the songs created with EasyPlay!, the songs created with the DAW, and the songs created by artists.

Further post-hoc tests or pairwise comparisons may be conducted to determine the specific differences between the categories. However, based on the initial findings, it can be concluded that songs created by artists received a significantly higher average rating compared to the other two categories.

Additionally, while the ANOVA results suggest a statistically significant difference, further analysis may be required to assess the specific contrast between the songs created with the software and the songs created with the DAW. It is worth noting that, although not statistically significant, there is a tendency for the songs created with the software to have higher ratings compared to those created with the basic DAW.

In summary, the ANOVA test supports the hypothesis that songs created by artists received a better overall rating. Furthermore, there is preliminary evidence to suggest that the songs created with our software have the potential to achieve higher ratings compared to those created with the DAW, although additional investigation may be warranted to confirm this observation.

E. Analysis Conclusion

The combined analysis of the qualitative and quantitative data provides a comprehensive understanding of the evaluation results. The qualitative analysis illuminated the participants' experiences and perceptions, highlighting the software's strengths in guiding non-musicians in the music creation process and supporting their creative engagement. The quantitative analysis further reinforced these findings, demonstrating the positive evaluations of the compositions created using the software.

The qualitative data and analysis indicate that the software successfully supports non-musicians' creative engagement while guiding them in the music creation process. The results support the proposition that the software can simplify the music creation process for non-musicians and provide a user-friendly interface for their creative exploration.

Findings in the quantitative analysis indicate a statistically significant difference between songs created by Easyplay!, Reaper and the local artists, but additional research may be necessary to fully understand the differences between the songs. Although not statistically significant, it is important to note that songs produced using EasyPlay! typically receive higher ratings than songs produced using the default DAW.

VI. CONCLUSION

In this project, we set out to explore how a music production software design can guide and support non-musicians' creative engagement. The research question aimed to understand the potential of a new software design in simplifying the music creation process and empowering non-musicians to express their creativity.

Through a comprehensive exploration of various fields such as Digital Audio Workstations, Music and Emotions, and Creative Engagement, a music production software prototype that aimed to address the needs of non-musicians is developed. The software design consisted of generators and a sequencer with graphical scores, all presented through a user-friendly interface following a flat design approach.

To ensure the effectiveness of our software design, we conducted a survey to determine user preferences for different prototypes and their associations with music and color. The results of this survey provided valuable insights into the preferences and expectations of our target users.

Furthermore, we conducted both qualitative and quantitative analyses to evaluate the impact and effectiveness of our software design. The qualitative analysis, based on semi-structured interviews, revealed that our software was perceived as a valuable tool for creative engagement, providing guidance and a sense of impact on the music creation process. Participants described it as a platform akin to playing with Legos, allowing them to explore different possibilities and experiment with musical ideas.

Additionally, the quantitative analysis, based on a questionnaire administered to music experts and students, provided further insights into the perceived creativity and usability of the software. While the results indicated that songs created by artists received the highest ratings overall, there was evidence to suggest that songs created with our software performed favorably compared to those created with the basic DAW, although this difference may not be statistically significant.

Overall, this project demonstrates the potential of a well-designed music production software in guiding and supporting non-musicians' creative engagement. By simplifying the music creation process and providing intuitive tools and features, non-musicians can express their creativity and explore the realm of music composition.

However, it is important to acknowledge that this project has its limitations. Further research and testing may be required to validate and refine the software design, ensuring its effectiveness and usability across a broader range of users. The use of Artificial Intelligence within the generator algorithms, the enhancement of the sample library size and a improvements in the fluidity and the quality of functionalities

might produce more significant results. Additionally, the software's integration into an existing open-source DAW opens up opportunities for future collaborations and improvements.

In conclusion, this project contributes to the understanding of how a music production software design can support non-musicians to engage creatively with music. By considering the complexity decrease process, incorporating key findings from related fields, and conducting thorough evaluations, we have developed a software design that holds promise in supporting non-musicians on their musical journey, fostering creativity, and enabling them to create their own expressive compositions.

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